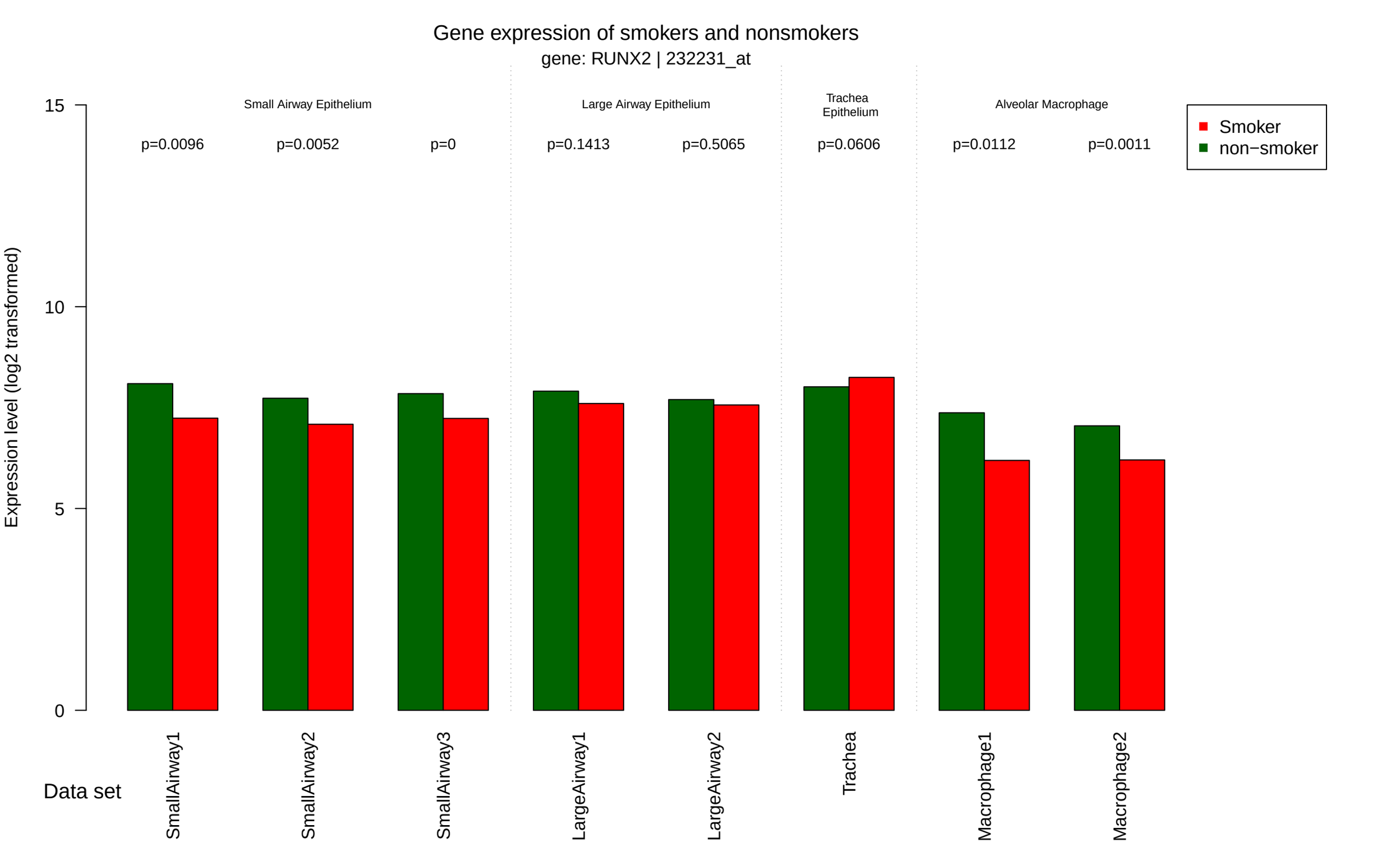
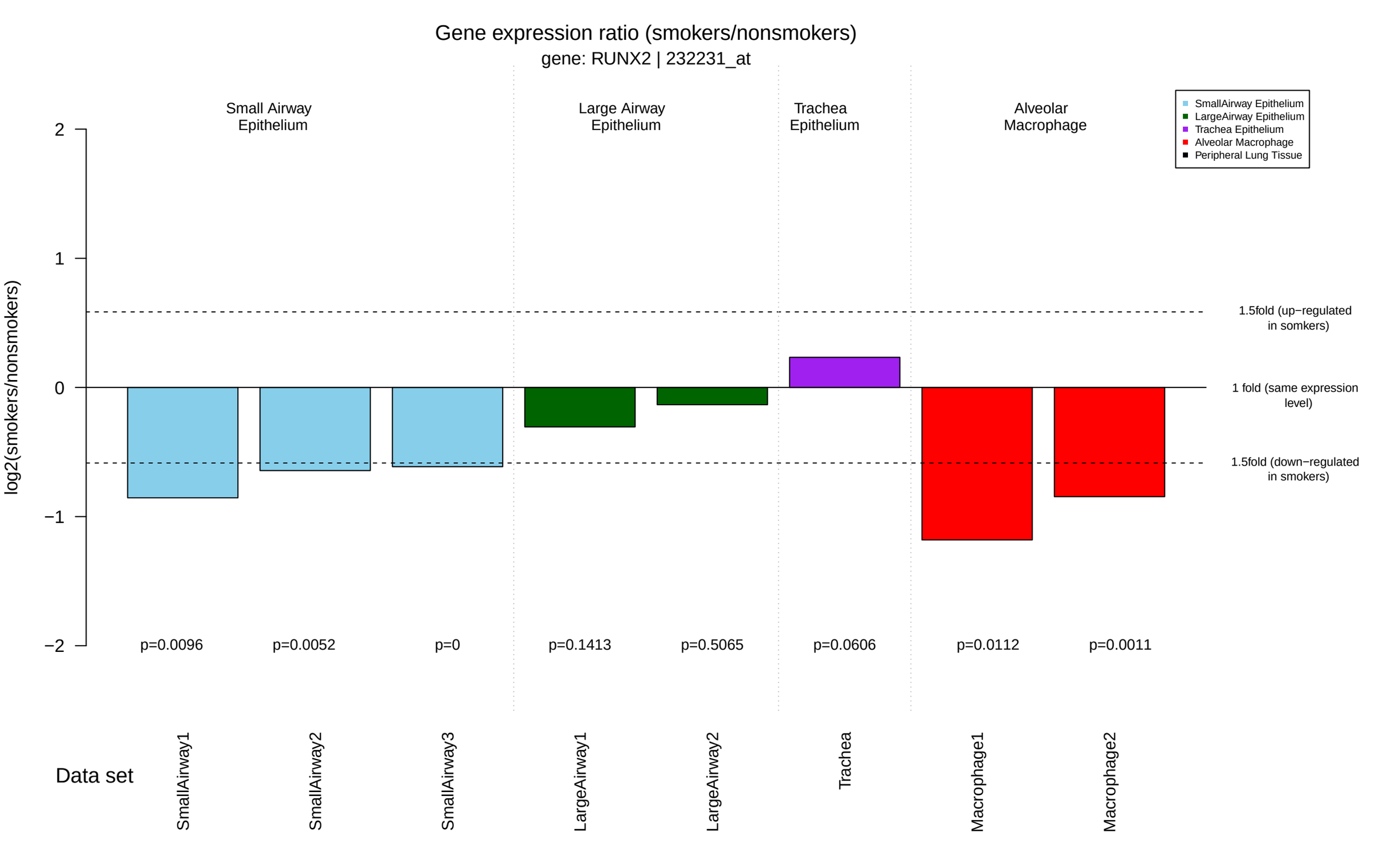
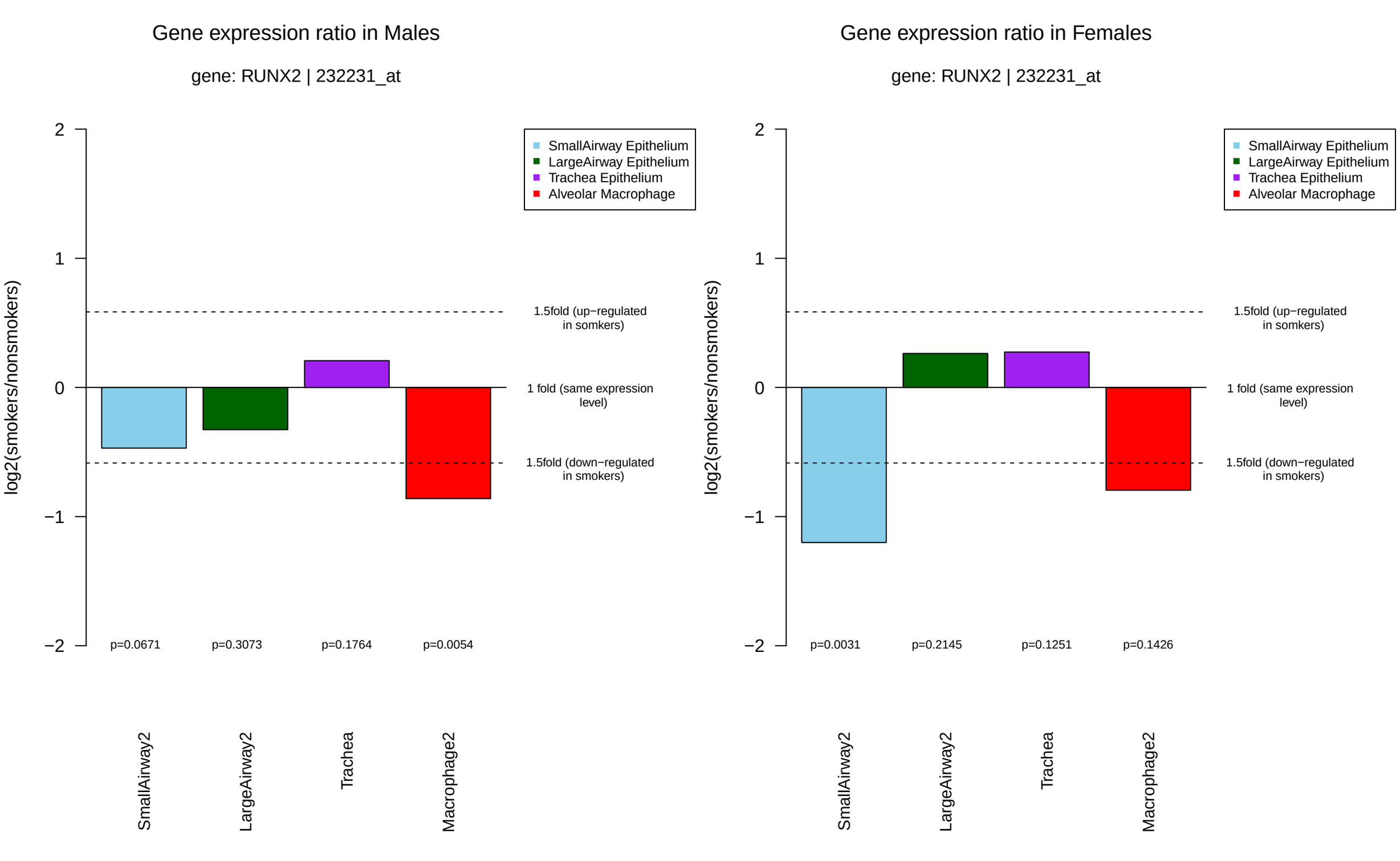
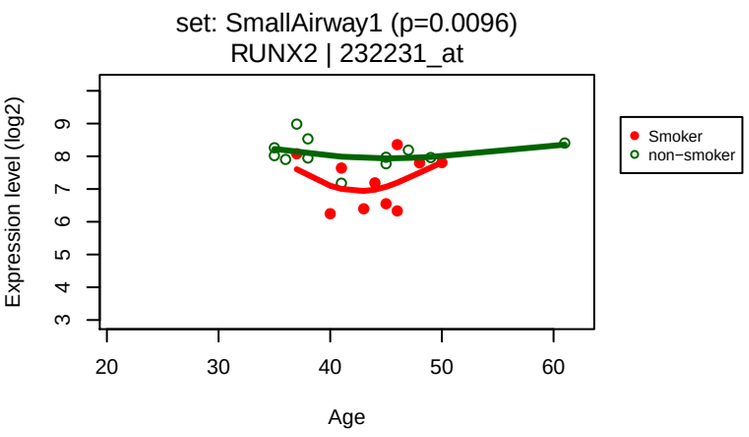
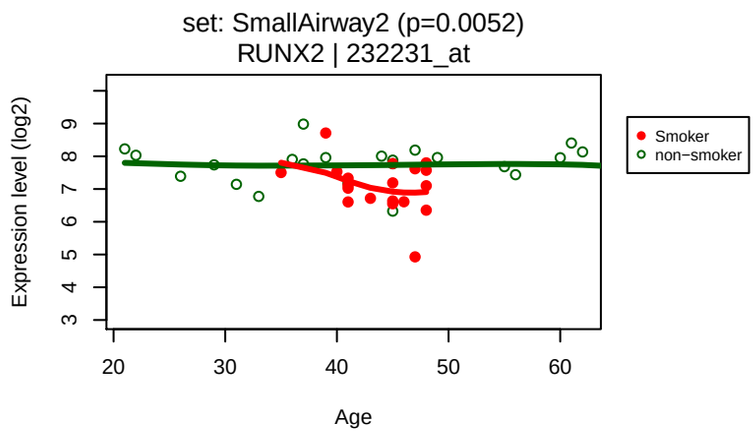
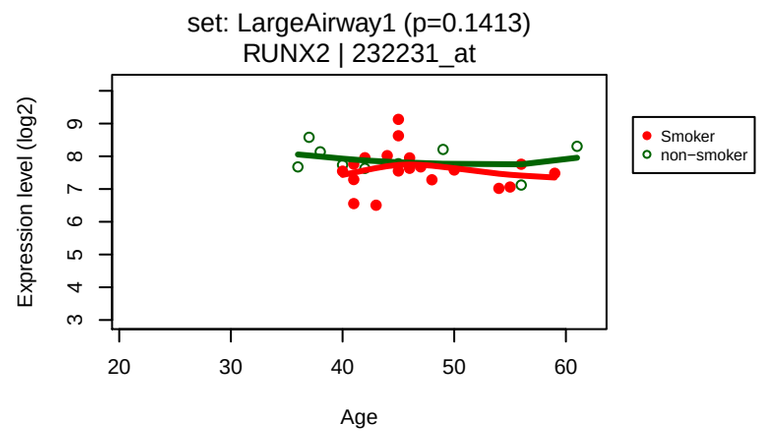

Gene expression of smokers and nonsmokers
gene: RUNX2 | 232231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0096
p=0.0052
p=0
p=0.1413
p=0.5065
p=0.0606
p=0.0112
p=0.0011
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

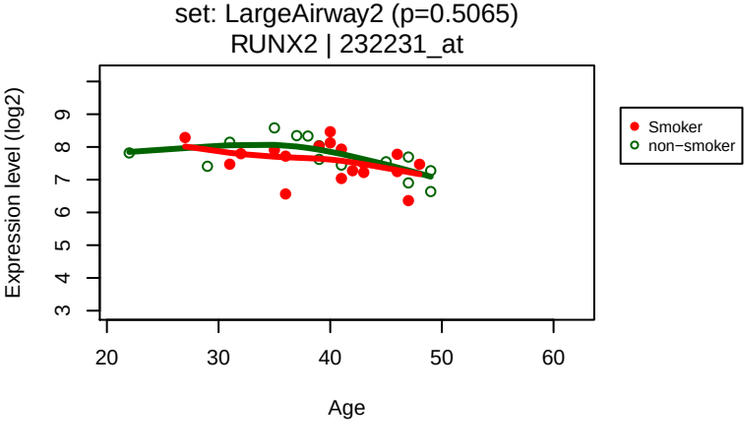

Gene expression ratio (smokers/nonsmokers)
gene: RUNX2 | 232231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0096
p=0.0052
p=0
p=0.1413
p=0.5065
p=0.0606
p=0.0112
p=0.0011
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

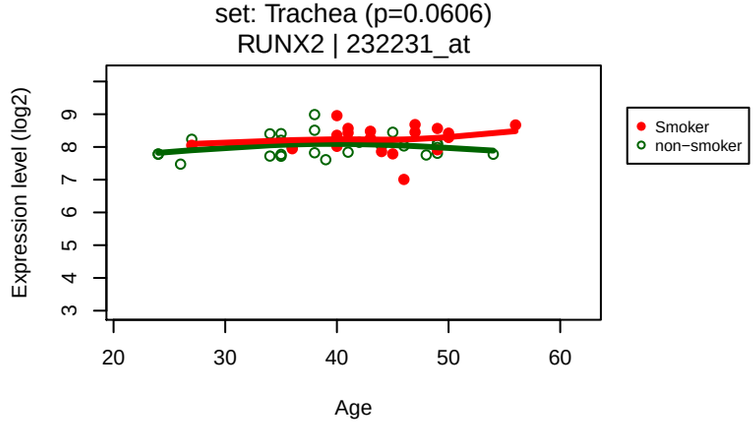

Gene expression ratio in Males
gene: RUNX2 | 232231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0671
p=0.3073
p=0.1764
p=0.0054
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX2 | 232231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.2145
p=0.1251
p=0.1426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

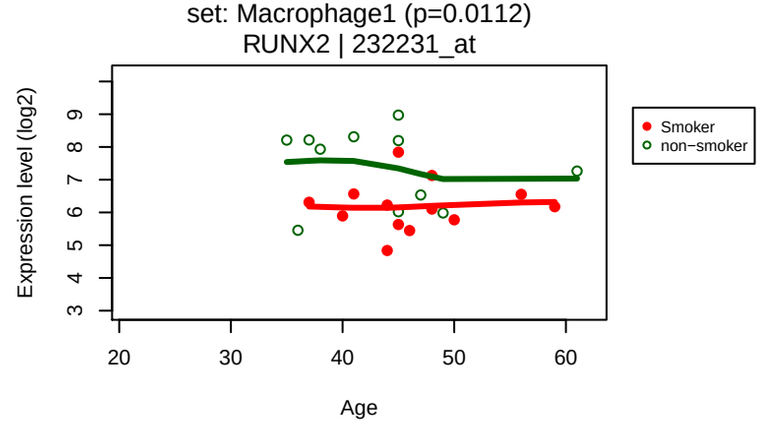

set: SmallAirway1 (p=0.0096)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker

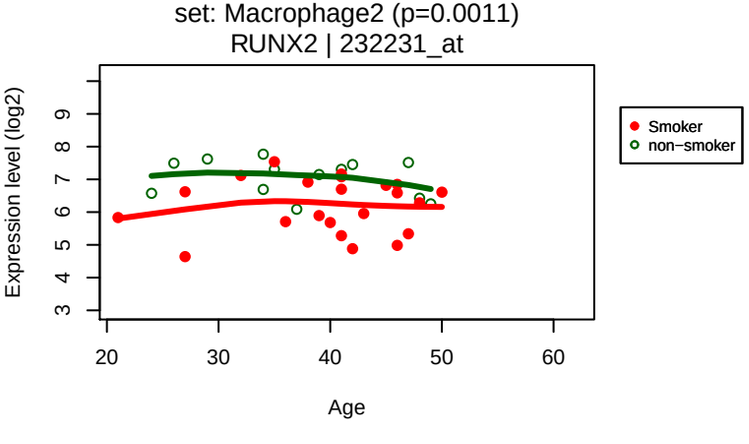

set: SmallAirway2 (p=0.0052)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker

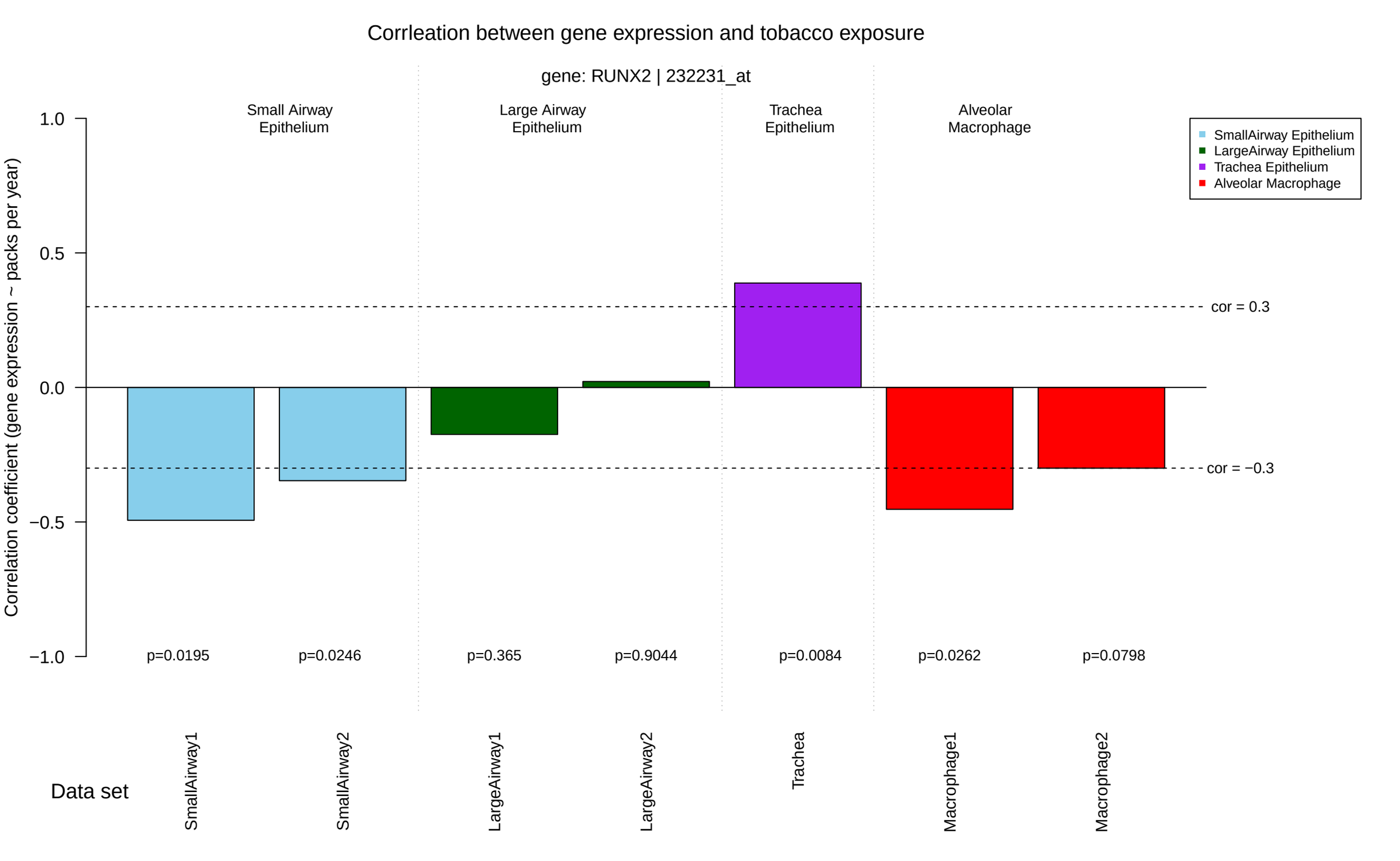

set: LargeAirway1 (p=0.1413)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5065)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0606)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0112)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0011)
RUNX2 | 232231_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX2 | 232231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0195
p=0.0246
p=0.365
p=0.9044
p=0.0084
p=0.0262
p=0.0798
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2